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== Vitis HLS Report for 'matVecMul'
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* Date: Tue May 5 15:46:25 2026

* Version: 2021.1 (Build 3247384 on Thu Jun 10 19:36:07 MDT 2021)

* Project: matVecMul_stream_prj

* Solution: solution_vanilla (Vivado IP Flow Target)

* Product family: zynq

* Target device: xc7z020-clg400-1

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== Performance Estimates
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+ Timing:

* Summary:

Clock	Target	Estimated	Uncertainty
ap_clk	10.00 ns	5.831 ns	2.70 ns

+ Latency:

* Summary:

Latency (cycles)		Latency (absolute)		Interval		Pipeline
min	max	min	max	min	max	Type
4872	9743	48.720 us	97.430 us	4873	9744	no

+ Detail:

* Instance:

Instance	Module	Latency (cycles)		Latency (absolute)		Interval		Pipeline
		min	max	min	max	min	max	Type
grp_matVecMul_Pipeline_sample_loop_fu_219	matVecMul_Pipeline_sample_loop	66	66	0.660 us	0.660 us	66	66	no
grp_matVecMul_Pipeline_matmul_inner_loop_fu_240	matVecMul_Pipeline_matmul_inner_loop	70	70	0.700 us	0.700 us	70	70	no

* Loop:

Loop Name	Latency (cycles)		Iteration	Initiation	Interval	Trip	
	min	max	Latency	achieved	target	Count	Pipelined
- vector_loop	4871	9742	4871	-	-	1 ~ 2	no
+ matmul_outer_loop	4800	4800	75	-	-	64	no

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== Utilization Estimates
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* Summary:

Name	BRAM_18K	DSP	FF	LUT	URAM
DSP	-	-	-	-	-
Expression	-	-	0	294	-
FIFO	-	-	-	-	-
Instance	4	1	392	673	-
Memory	1	-	0	0	0

Multiplexer	-	-	-	132	-
Register	-	-	279	-	-
Total	5	1	671	1099	0
Available	280	220	106400	53200	0
Utilization (%)	1	~0	~0	2	0

+ Detail:

* Instance:

Instance	Module	BRAM_18K	DSP	FF	LUT	URAM
control_s_axi_U	control_s_axi	4	0	230	302	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_240	matVecMul_Pipeline_matmul_inner_loop	0	1	153	299	0
grp_matVecMul_Pipeline_sample_loop_fu_219	matVecMul_Pipeline_sample_loop	0	0	9	72	0
Total		4	1	392	673	0

* DSP:

N/A

* Memory:

Memory	Module	BRAM_18K	FF	LUT	URAM	Words	Bits	Banks	W*Bits*Banks
x_localbuff_V_U	x_localbuff_V	1	0	0	0	64	16	1	1024
Total		1	0	0	0	64	16	1	1024

* FIFO:

N/A

* Expression:

Variable Name	Operation	DSP	FF	LUT	Bitwidth P0	Bitwidth P1
add_ln32_fu_274_p2	+	0	0	38	31	1
add_ln44_fu_285_p2	+	0	0	39	32	1
p_Val2_s_fu_353_p2	+	0	0	23	16	16
sub_fu_251_p2	+	0	0	39	32	2
and_ln789_fu_439_p2	and	0	0	2	1	1
and_ln790_fu_453_p2	and	0	0	2	1	1
and_ln795_fu_483_p2	and	0	0	2	1	1
carry_1_fu_373_p2	and	0	0	2	1	1
overflow_fu_477_p2	and	0	0	2	1	1
underflow_fu_501_p2	and	0	0	2	1	1
Range1_all_ones_fu_405_p2	icmp	0	0	12	14	2
Range1_all_zeros_fu_411_p2	icmp	0	0	12	14	1
Range2_all_ones_fu_389_p2	icmp	0	0	12	13	2
icmp_ln32_fu_269_p2	icmp	0	0	18	32	32
icmp_ln44_fu_280_p2	icmp	0	0	18	32	32
out_pkt_last_V_fu_295_p2	icmp	0	0	18	32	32
or_ln384_fu_515_p2	or	0	0	2	1	1
or_ln794_fu_465_p2	or	0	0	2	1	1
or_ln795_fu_489_p2	or	0	0	2	1	1
deleted_ones_fu_445_p3	select	0	0	2	1	1
deleted_zeros_fu_417_p3	select	0	0	2	1	1
select_ln384_fu_507_p3	select	0	0	17	1	15
val_V_fu_521_p3	select	0	0	16	1	16
xor_ln416_fu_367_p2	xor	0	0	2	1	2
xor_ln789_fu_433_p2	xor	0	0	2	1	2
xor_ln794_1_fu_471_p2	xor	0	0	2	1	2

xor_ln794_fu_459_p2	xor	0	0	2	1	2
xor_ln795_fu_495_p2	xor	0	0	2	1	2
Total		0	0	294	266	173

* Multiplexer:

Name	LUT	Input Size	Bits	Total Bits
ap_NS_fsm	59	11	1	11
i_fu_152	9	2	31	62
in_stream_TREADY_int_regslice	9	2	1	2
k_reg_208	9	2	32	64
out_stream_TDATA_blk_n	9	2	1	2
x_localbuff_V_address0	14	3	6	18
x_localbuff_V_ce0	14	3	1	3
x_localbuff_V_we0	9	2	1	2
Total	132	27	74	164

* Register:

Name	FF	LUT	Bits	Const Bits
add_ln32_reg_571	31	0	31	0
add_ln44_reg_579	32	0	32	0
ap_CS_fsm	10	0	10	0
col_size_read_reg_541	32	0	32	0
grp_matVecMul_Pipeline_matmul_inner_loop_fu_240_ap_start_reg	1	0	1	0
grp_matVecMul_Pipeline_sample_loop_fu_219_ap_start_reg	1	0	1	0
i_fu_152	31	0	31	0
k_reg_208	32	0	32	0
no_of_vectors_read_reg_552	32	0	32	0
out_pkt_last_V_reg_589	1	0	1	0
row_size_read_reg_547	32	0	32	0
shl_ln_reg_594	6	0	13	7
sub_reg_563	32	0	32	0
trunc_ln1169_reg_584	6	0	6	0
Total	279	0	286	7

== Interface

* Summary:

RTL Ports	Dir	Bits	Protocol	Source Object	C Type
s_axi_control_AWVALID	in	1	s_axi	control	array
s_axi_control_AWREADY	out	1	s_axi	control	array
s_axi_control_AWADDR	in	14	s_axi	control	array
s_axi_control_WVALID	in	1	s_axi	control	array
s_axi_control_WREADY	out	1	s_axi	control	array
s_axi_control_WDATA	in	32	s_axi	control	array
s_axi_control_WSTRB	in	4	s_axi	control	array
s_axi_control_ARVALID	in	1	s_axi	control	array
s_axi_control_ARREADY	out	1	s_axi	control	array
s_axi_control_ARADDR	in	14	s_axi	control	array
s_axi_control_RVALID	out	1	s_axi	control	array
s_axi_control_RREADY	in	1	s_axi	control	array
s_axi_control_RDATA	out	32	s_axi	control	array
s_axi_control_RRESP	out	2	s_axi	control	array

s_axi_control_BVALID	out	1	s_axi	control	array
s_axi_control_BREADY	in	1	s_axi	control	array
s_axi_control_BRESP	out	2	s_axi	control	array
ap_clk	in	1	ap_ctrl_hs	matVecMul	return value
ap_rst_n	in	1	ap_ctrl_hs	matVecMul	return value
interrupt	out	1	ap_ctrl_hs	matVecMul	return value
in_stream_TDATA	in	32	axis	in_stream_V_data_V	pointer
in_stream_TVALID	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TREADY	out	1	axis	in_stream_V_dest_V	pointer
in_stream_TDEST	in	1	axis	in_stream_V_dest_V	pointer
in_stream_TKEEP	in	4	axis	in_stream_V_keep_V	pointer
in_stream_TSTRB	in	4	axis	in_stream_V_strb_V	pointer
in_stream_TUSER	in	1	axis	in_stream_V_user_V	pointer
in_stream_TLAST	in	1	axis	in_stream_V_last_V	pointer
in_stream_TID	in	1	axis	in_stream_V_id_V	pointer
out_stream_TDATA	out	32	axis	out_stream_V_data_V	pointer
out_stream_TVALID	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TREADY	in	1	axis	out_stream_V_dest_V	pointer
out_stream_TDEST	out	1	axis	out_stream_V_dest_V	pointer
out_stream_TKEEP	out	4	axis	out_stream_V_keep_V	pointer
out_stream_TSTRB	out	4	axis	out_stream_V_strb_V	pointer
out_stream_TUSER	out	1	axis	out_stream_V_user_V	pointer
out_stream_TLAST	out	1	axis	out_stream_V_last_V	pointer
out_stream_TID	out	1	axis	out_stream_V_id_V	pointer